

Tsu-Jae King Liu

其他

基本信息

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地区: United States

学科领域

微电子科学与工程

集成电路设计与集成系统

研究领域

微电子器件与制造

纳米电子与先进工艺

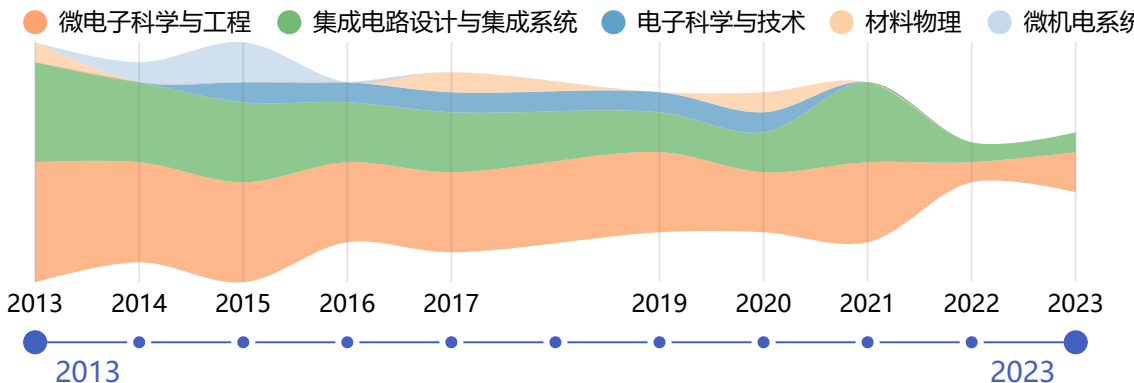
低功耗电路与能量管理

存储器设计与架构

个人简介

Tsu-Jae King Liu is the Dean of the College of Engineering at the University of California, Berkeley. She is based at McLaughlin Hall and can be reached at +1 510-642-5771. Her department is ED1DO - Col of Engin Dean's Office.

近年发表的论文



Experimental Demonstration of Coupled Sub-Harmonic Injection Locked Oscillation in Micro-Electro-Mechanical Relays

IEEE ELECTRON DEVICE LETTERS

Hu, Xiaoler; Tatum, Lars Prospero; Almeida, Sergio Fabian; Esatu, Tsegereda Kedir; Liu, Tsu-Jae King

被引用次数: 0

2023

Non-Volatile Nano-Electro-Mechanical Switches and Hybrid Circuits in a 16 nm CMOS Back-End-of-Line Process

IEEE ELECTRON DEVICE LETTERS

Sikder, Urmita; Naous, Rawan; Stojanovic, Vladimir; Liu, Tsu-Jae King

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2023

Highly Reliable and Secure PUF Using Resistive Memory Integrated Into a 28 nm CMOS Process

IEEE TRANSACTIONS ON ELECTRON DEVICES

Esatu, Tsegereda K.; Prakash, Amit; Li, Zhi; Lau, Derek; Jo, Sung Hyun; Liu, Tsu-Jae King

被引用次数: 2

2023

Breakdown and Healing of Tungsten-Oxide Films on Microelectromechanical Relay Contacts

JOURNAL OF MICROELECTROMECHANICAL SYSTEMS

Falicov, Ethan; Marvin, Jessica; Ye, Zhixin Alice; Almeida, Sergio Fabian; Contreras, Daniel S.; King Liu, Tsu-Jae; Spencer, Matthew

被引用次数: 1

2022

Simulation-Based Study of Low Minimum Operating Voltage SRAM With Inserted-Oxide FinFETs and Gate-All-Around Transistors

IEEE TRANSACTIONS ON ELECTRON DEVICES

Wu, Yi-Ting; Ding, Fei; Chiang, Meng-Hsueh; Chen, Jone F.; Liu, Tsu-Jae King

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Design Technology Co-Optimization for Back-End-of-Line Nonvolatile NEM Switch Arrays

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2021

Toward Monolithically Integrated Hybrid CMOS-NEM Circuits

IEEE TRANSACTIONS ON ELECTRON DEVICES

Sikder, Urmita; Horace-Herron, Kelsey; Yen, Ting-Ta; Usai, Giulia; Hutin, Louis; Stojanovic, Vladimir; Liu, Tsu-Jae King

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Body-Biased Multiple-Gate Micro-Electro-Mechanical Relays

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被引用次数: 7

2020

Sublithographic Patterning of Spin-Coated SiARC Films Using Tilted Ion Implantation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Rembert, Thomas R.; Sharma, Shalini; Garcia, Luis; Connelly, Daniel; Tomoya, Taji; Sakai, Tatsuya; Rubin, Leonard; Liu, Tsu-Jae King

被引用次数: 0

2020

A Density Metric for Semiconductor Technology

PROCEEDINGS OF THE IEEE

Wong, H. -S. Philip; Akarvardar, Kerem; Antoniadis, Dimitri; Bokor, Jeffrey; Hu, Chenming; King-Liu, Tsu-Jae; Mitra, Subhasish; Plummer, James D.; Salahuddin, Sayeef

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Sikder, Urmita; Usai, Giulia; Yen, Ting-Ta; Horace-Herron, Kelsey; Hutin, Louis; Liu, Tsu-Jae King

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ZrO₂ Ferroelectric FET for Non-volatile Memory Application

IEEE ELECTRON DEVICE LETTERS

Liu, Huan; Wang, Chengxu; Han, Genquan; Li, Jing; Peng, Yue; Liu, Yan; Wang, Xingsheng; Zhong, Ni; Duan, Chungang; Wang, Xinran; Xu, Nuo; Liu, Tsu-Jae King; Hao, Yue

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2019

Simulation-Based Study of High-Density SRAM Voltage Scaling Enabled by Inserted-Oxide FinFET Technology

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Nanocrystal-Embedded-Insulator (NEI) Ferroelectric Field-Effect Transistor Featuring Low Operating Voltages and Improved Synaptic Behavior

IEEE ELECTRON DEVICE LETTERS

Peng, Yue; Xiao, Wenwu; Han, Genquan; Liu, Yan; Wu, Jibao; Wang, Kuan; He, Yuhui; Yu, Zhihao; Wang, Xinran; Xu, Nuo; Liu, Tsu-Jae King; Hao, Yue

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Simulation-Based Study of Si/Si_{0.9}Ge_{0.1}/Si Hetero-Channel FinFET for Enhanced Performance in Low-Power Applications

IEEE ELECTRON DEVICE LETTERS

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High-Mobility Ge pMOSFETs With Crystalline ZrO₂ Dielectric

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Liu, Huan; Han, Genquan; Xu, Yang; Liu, Yan; Liu, Tsu-Jae King; Hao, Yue

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Nanocrystal-Embedded-Insulator Ferroelectric Negative Capacitance FETs With Sub-kT/q Swing

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Variability Study for Low-Voltage Microelectromechanical Relay Operation

IEEE TRANSACTIONS ON ELECTRON DEVICES

Osoba, Benjamin; Saha, Bivas; Almeida, Sergio F.; Patil, Jatin; Brandt, Laura E.; Roots, Maurice E. D.; Acosta, Edgar; Wu, Junqiao; Liu, Tsu-Jae King

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2018

Channel Stress and Ballistic Performance Advantages of Gate-All-Around FETs and Inserted-Oxide FinFETs

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Connelly, Daniel; Zheng, Peng; Liu, Tsu-Jae King

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2017

Comparison of SOI Versus Bulk FinFET Technologies for 6T-SRAM Voltage Scaling at the 7-/8-nm Node

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Zhang, Xi; Connelly, Daniel; Takeuchi, Hideki; Hytha, Marek; Mears, Robert J.; Liu, Tsu-Jae King

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Reducing adhesion energy of micro-relay electrodes by ion beam synthesized oxide nanolayers

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Modeling Nanoelectromechanical Switches With Random Surface Roughness

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Non-Volatile Nano-Electro-Mechanical Memory for Energy-Efficient Data Searching

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Kato, Kimihiko; Stojanovic, Vladimir; Liu, Tsu-Jae King

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Demonstration of L-Shaped Tunnel Field-Effect Transistors

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Kato, Kimihiko; Stojanovic, Vladimir; Liu, Tsu-Jae King

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Nanomechanical Switch Designs to Overcome the Surface Adhesion Energy Limit

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Effect of Body Biasing on the Energy-Delay Performance of Logic Relays

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Threshold Voltage and DIBL Variability Modeling Based on Forward and Reverse Measurements for SRAM and Analog MOSFETs

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Damrongplasit, Nattapol; Zamudio, Luis; Liu, Tsu-Jae King; Balasubramanian, Sriram

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Simulation-Based Study of the Inserted-Oxide FinFET for Future Low-Power System-on-Chip Applications

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Zheng, Peng; Connelly, Daniel; Ding, Fei; Liu, Tsu-Jae King

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2015

Germanium Gate PhotoMOSFET Integrated to Silicon Photonics

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Going, Ryan W.; Loo, Jodi; Liu, Tsu-Jae King; Wu, Ming C.

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Adhesive Force Characterization for MEM Logic Relays With Sub-Micron Contacting Regions

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Yaung, Jack; Hutin, Louis; Jeon, Jaeseok; Liu, Tsu-Jae King

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Electromechanical Diode Cell Scaling for High-Density Nonvolatile Memory

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Hutin, Louis; Kwon, Wookhyun; Qian, Chuang; Liu, Tsu-Jae King

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Design of Gate-All-Around Silicon MOSFETs for 6-T SRAM Area Efficiency and Yield

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Extension of Planar Bulk n-Channel MOSFET Scaling With Oxygen Insertion Technology

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Microelectromechanical Relay and Logic Circuit Design for Zero Crowbar Current

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Fujiki, Jun; Xu, Nuo; Hutin, Louis; Chen, I-Ru; Qian, Chuang; Liu, Tsu-Jae King

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Variation-Aware Comparative Study of 10-nm GAA Versus FinFET 6-T SRAM Performance and Yield

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Comparative Study of Uniform Versus Supersteep Retrograde MOSFET Channel Doping and Implications for 6-T SRAM Yield

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Raised-Source/Drain Double-Gate Transistor Design Optimization for Low Operating Power

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Design Optimization of Multigate Bulk MOSFETs

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Fabrication of Si_{1-x}Gex/Si pMOSFETs Using Corrugated Substrates for Improved ION and Reduced Layout-Width Dependence

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Study of Random Dopant Fluctuation Induced Variability in the Raised-Ge-Source TFET

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A New Switching Device for Printed Electronics: Inkjet-Printed Microelectromechanical Relay

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Park, Eung Seok; Chen, Yenhao; Liu, Tsu-Jae King; Subramanian, Vivek

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Mechanically modulated tunneling resistance in monolayer MoS₂

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Impact of Gate Line-Edge Roughness (LER) Versus Random Dopant Fluctuations (RDF) on Germanium-Source Tunnel FET Performance

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Damrongplasit, Nattapol; Kim, Sung Hwan; Shin, Changhwan; Liu, Tsu-Jae King

被引用次数: 15

2013

学术合作



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